

Topic of the Book	:	Pedagogy of Biological Science - I
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Edition	:	First Edition – Jan. 2022
Copy right to	:	Everest Publishers
Publisher	:	Everest Publishers, Chennai – 26.
Printed at	:	S.K.S. Printers, Chennai - 2
Price	:	₹ 250

Copies can available at the following address.

EVEREST PUBLISHERS

Pearl Nest Apartment,

5/A, Thirunagar 1st Street,
Vadapalani,

Chennai – 600 026.

Cell No : 9894472790 / 044-42070507

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J.K. Garden First Street,

Ukkadam,

Coimbatore – 641 008.

Cell No : 9894472790 / 9488551302

PREFACE

Pedegogy of Biological Science - I is a book prepared under the 2021 - 22 syllabus of TNTE University for the B.Ed., First Semester Students. Thou this book is in use for only for one semester to B.Ed., students, I prepared it for teacher educators and students who opted for English medium. The main aim of publishing this book is to make the students to learn teaching of Biological Science with ease and pass the examination in English Medium.

There are so many books available on the subject written by worthy authors. However every writer has his own style and way of presentation. This book has its own features and characteristics. Biological Science concepts are well illustrated with suitable examples and diagrams. I thank almighty God who helps me to bring this book in a short period.

In writing this book, I have used various ideas and works of eminent writers and from Google network. I express my sincere thanks to all the writers in completing this book.

Suggestion, Comments from teacher educators, teacher trainees shall be appreciated.

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Chennai,
02.01.2022

PEDAGOGY OF BIOLOGICAL SCIENCE - I

University Syllabus 2021-2022 onwards

UNIT- I: AIMS AND OBJECTIVES OF TEACHING BIOLOGICAL SCIENCE

Biological Science: Meaning -Aims and objectives of teaching Biological Science in schools – Need and significance of teaching Biological Science- Nature – Scope -Values of Teaching Biological Science. Bloom's Taxonomy of Instructional Objectives: Cognitive, Affective and Psychomotor Domains, Revised Bloom's Taxonomy 2001 (Anderson & Krathwohl) Interrelation among the domains – Correlation between subjects.

UNIT-II: TEACHING SKILLS

Micro-Teaching: Concept, Definition, Steps and Cycle - Micro-teaching Vs Macro-Teaching - Skill of Set Induction - Skill of Explaining, Skill of Questioning, Probing skills, Skill of Stimulus Variation, Skill of Reinforcement, Skill of non-verbal clues, Skill of Closure, Skill of Black Board Usage - Link lesson – Model episode.

UNIT – III: APPROACHES OF TEACHING

Approaches of Teaching Biological Science: The Concentric Approach, Topical Approach, Chronological Approach, Unit Approach, Correlated Approach and Integrated Approach - Lesson Planning: Need for Lesson Planning, Steps in Lesson Planning, - Organizing Teaching: Memory Level (Herbartian Model), Understanding Level (Morrison teaching Model), Reflective Level (Bigge and Hunt Teaching Model)– Unit Plan – Lesson Plan Writing.

UNIT-IV: METHODS OF TEACHING

Teacher-centred methods: Lecture method – Demonstration method – Team Teaching. Learner –centred methods: Laboratory method - Peer tutoring/ teaching by students – Project method – Individual activities – Experiential method – Teacher guided learning – Problem~ solving method –Small group/whole class interactive learning – Students' Seminar – Group discussion. Recent Trends: Constructivist learning – Problem-based learning – Brain-based learning – Collaborative learning.

UNIT-V: INSTRUCTIONAL MEDIA

Print Resources: Newspapers – Journals and magazines – Science Encyclopaedias. Audio Resources: Radio talk – Audio Tapes – DVDs/CDs. Visual resources: Pictures – Flash cards – charts – Posters – Photographs – Models. ICT Resources: Radio – Television- Internet, Multimedia, Interactive whiteboard, Online Teaching Resources. Community resources: Zoological gardens, Botanical gardens, Eco-park- Aquarium – Science Exhibition / Fair – Fieldtrip –New Emerging Media: Tele-Conferencing, Communication Satellites, Computer Networking, Word Processors, Blended Learning, Flipped Classroom, Artificial Intelligence and Augmented Reality - Qualities of a good Biology Textbook – Qualities of a Biology teacher.

PEDAGOGY OF BIOLOGICAL SCIENCE

CONTENT

Unit	Topic	Page Nos.
1.	Aims and Objectives of Biological Science	
	1.1 Biological Science - Meaning	11
	1.2 Aims and Objectives of Teaching Biological Science in schools	12
	1.2.1 Aims of teaching Biological Science in Schools	14
	1.2.2 Difference between Aims and Objectives	18
	1.3 Need and Significance Teaching Biological Science	19
	1.4 Nature of Biological Science	23
	1.5 Scope of Biological Science	26
	1.6 Values of Teaching Biological Science	27
	1.7 Blooms Taxonomy of Instructional Objectives	30
	1.7.1 Cognitive Domain	30
	1.7.2 Affective Domain	35
	1.7.3 Psychomotor Domain	37
	1.8 Revised Bloom's Taxonomy	40
	1.9 Writing Instructional for Different Biological Topics	43
	1.10 Inter-relation among the domains	47
	1.11 Correlation between Biological Science and other subjects	49
2.	Teaching skills	
	2.1 Micro Teaching	55
	2.1.1. Micro Teaching - Definition	55
	2.1.2. Need and Importance of Micro teaching	56
	2.1.3. Characteristics of Micro teaching	56

Unit	Topic	Page Nos.
	2.1.4. Micro Teaching cycle	57
	2.1.5. Uses of Micro Teaching	58
	2.1.6. Micro Teaching Vs. Macro Teaching	59
	2.2 Identification of teaching skills	61
	2.2.1. Skill of Set induction	63
	2.2.2. Skill of Explaining	65
	2.2.3. Skill of questioning	67
	2.2.4. Skill of probing	69
	2.2.5. Skill of Stimulus variation	70
	2.2.6. Skill of Reinforcement	72
	2.2.7. Skill of Non-verbal cues	74
	2.2.8. Skill of closure	76
	2.2.9. Skill of Blackboard Usage	77
	2.2.10. Advantages of micro teaching	79
	2.3 Model Micro Lessons	80
	2.4 Link Lesson	84
	2.5 Model Episode	93
3.	Approaches of Teaching	
	3.1 Approaches of Teaching Biological Science	97
	3.1.1. The Concentric Approach	97
	3.1.2. Topical Approach	99
	3.1.3. Unit Approach	101
	3.1.4. Correlated Approach	103
	3.1.5. Integrated Approach	107
	3.2 Steps in planning a lesson	109
	3.3 Designing a Lesson Plan	111
	3.2.1. Herbartian Steps	114
	3.2.2. Preparation of Lesson Plan	115
	3.4 Organising Teaching	117
	3.4.1 Memory Level of Teaching	118
	3.4.2 Understanding Level of Teaching	123
	3.4.3 Reflective Level of Teaching	126
	3.5 Designing a Unit plan	129

Unit	Topic	Page Nos.
	3.5.1 Difference between Lesson plan and unit plan	135
	3.6 Structure of a four-fold lesson plan	135
	3.7 Preparation of Model Lesson Plans	140
4.	Methods of Teaching	
	4.1. Teacher-centered methods	151
	4.1.1 Lecture Method	151
	4.1.2 Demonstration Method	154
	4.1.3 Team Teaching	158
	4.2. Learner-centered methods	160
	4.2.1 Laboratory Method	161
	4.2.2 Peer tutoring	163
	4.2.3 Project Method	165
	4.2.4 Individual Activities	169
	4.2.5 Experimental Method	171
	4.2.6 Teacher Guided Learning	172
	4.2.7 Problem solving method	175
	4.3. Small Group / Whole Class interactive learning	179
	4.3.1 Student Seminar	179
	4.3.2 Group Discussion	181
	4.4. Recent trends in teaching	182
	4.4.1 Constructivist Learning	182
	4.4.2 Problem based Learning	185
	4.4.3 Brain based Learning	187
	4.4.4 Collaborative Learning	190
5.	Instructional Media	
	5.1. Print resources:	193
	5.1.1 Newspapers	193
	5.1.2 Journals	195
	5.1.3 Magazines	197
	5.1.4 Science Encyclopaedias	198
	5.2. Audio resources	200
	5.2.1 Radio Talk	200
	5.2.2 Audio Tape	201

Unit	Topic	Page Nos.
	5.2.3 DVDs / CDs	203
	5.3. Visual resources	204
	5.3.1 Pictures	204
	5.3.2 Flash Cards	205
	5.3.3 Charts	207
	5.3.4 Posters	208
	5.3.5 Photographs	209
	5.3.6 Models	211
	5.4. ICT resources:	212
	5.4.1 Radio	212
	5.4.2 Television	214
	5.4.3 Internet	216
	5.4.4 Multimedia	219
	5.4.5 Interactive whiteboard	224
	5.4.6 Online Teaching Resources	225
	5.5. Community resources:	228
	5.5.1 Zoological Gardens	228
	5.5.2 Botanical Gardens	229
	5.5.3 Eco Park	231
	5.5.4 Aquarium	232
	5.5.5 Science Exhibition / Fair	233
	5.5.6 Field trips	237
	5.6 New Emerging Media	240
	5.6.1 Tele-conferencing	241
	5.6.2 Communication Satellites	244
	5.6.3 Computer Networking	246
	5.6.4 Word Processors	250
	5.6.5 Blended Learning	253
	5.6.6 Flipped Classroom	256
	5.6.7 Artificial Intelligence	260
	5.6.8 Augmented Reality	264
	5.7 Qualities of a good Biology text book	266
	5.8 Qualities of a Biology Teacher	269

UNIT - 1

AIMS AND OBJECTIVES OF TEACHING BIOLOGICAL SCIENCE

1.1 Biological Science : Meaning

Biology is an important part of the science. It includes zoology, botany and environmental science.

- It gives us the knowledge about the plants, animals , their habitats and also their relation with their environment.
- Environmental education is a branch of Biological, which gives us awareness about ecological balance, food web ,food chain and also the man made disorders in the ecosystem and their effects.
- Nature gives us the warning that if we are ecofriendly to nature, we can live in harmony. This is the lesson we can learn from Biology.
- Biology gives us the exact knowledge about all the living creatures in the world.
- Biology gives us the knowledge that nature is the only source which fulfills all the needs of plants and human beings.
- Biology train us to maintain nature and living organisms, in creating new breeds, research on living organisms and to find out new drugs.

The word science is derived from the Latin Word 'Scientia' which means 'knowledge'. Various scientists gave various explanations to science.

Science is the intellectual and practical activity encompassing the systematic study of the structure and behavior of the Biological and natural world through observation and experiment.

Science involves more than the gaining of knowledge. It is the systematic and organized inquiry into the natural world and its Phenomena.

Science is about gaining a deeper and often useful understanding of the world.

Science is Systematic knowledge of the Biological or material world gained through observation and experimentation.

1.2 Aims and objectives of teaching Biological Science in schools

Aims:

To teach any subject, it is the primary fact to know the teachers and students, what is the purpose of teaching and what are the uses of teaching. We use the term 'aims' to mean the broad goals which our educational system (the whole of the Indian society involved) embraces and which we are expected to attain. The aims of education are based on philosophical and socio-psychological aspects of society and culture. Some of the aims are as follows.

- Education to develop patriotism and good citizenship.
- Education for moral, ethical and spiritual values.

- Education to fulfill the vocational needs of the individual and society.
- Educational for leisure-time activities such as music, dancing, gardening, reading, hobbies, etc.,
- Education for improved health knowledge (Physical and mental) and practices.
- Special education for the disabled.

In our educational system, Biology is a compulsory subject from grade VI to grade X. Student who completed grade X should attain some basic concepts and accordingly behavioral changes are expected from the students. The aims and objectives differ a bit at different stages. Preliminary knowledge of objectives is required at early stages while complete and complex objectives are needed at higher stages. So capabilities of pupils should be kept in mind.

The aims of teaching Biology are different for different levels of education.

Accordingly, the content of Biology are fixed to their level which help them to gain behavioral changes.

Objectives:

Objectives are the milestones to reach the destination i.e. to attain the aim or ultimate goal of education. Objectives are specific, direct and practical in nature; objectives are related to the learning outcomes or change in the behaviour of the students. The objectives are meant for both the teacher and the students. Even at the planning stage i.e. before entering the classroom the teacher asks himself, "what changes in behaviour of the students can I bring about

through this lesson?" such changes constitute his (teacher's) educational objectives.

An objectives is thus

- A point or an end-view of something towards which action is directed;
- A planned change sought through an activity
- What we set out to do.

Educational objectives, therefore, consist of the changes we wish to produce in the child in the child. The changes that take place through education can be represented in

- The knowledge children acquire (knowledge)
- The skills and abilities children attain (skill)
- The interest children develop (interest) and
- The attitude children manifest (attitude)

He would understand something which he did not understand before. He can solve problems which he could not solve before. He favorably revises his attitudes towards things. Education is thus the process of bringing about changes in the behaviour of the students in

- What they know (knowledge extends)
- How they think (process of thinking changes)
- How they feel (process of feeling changes)
- How they work (methods of working change)

1.2.1 Aims of teaching Biological Science in schools

1. Concepts, information and skills which could be immediately used and applied.

2. Training for developing skills in handling effectively equipment and tools and also in the skill of problem solving.
3. Develop Biological science related attitudes.
4. Appreciating the contribution of Biological scientists and their inventions.

Training in making best use of leisure time and achieve knowledge and skills required for future life and progress.

Objectives of teaching Biological Science in schools

Are Successful Professionals

- Obtain employment in the Biological science field
- Gain acceptance into graduate school

Have a broad knowledge

- A broad knowledge of the theoretical foundations of Biological science
- Knowledge of the fundamental areas of Biological science
- Substantial knowledge of one key area of Biological science

Think Independently, Acquire knowledge and continue their development

- Acquire new knowledge
- Think independently and rigorously
- Develop a plan for professional development

Apply Scientific and Engineering Methodology to the Design, Implementation, Analysis and Evaluation

- Effectively apply scientific and engineering principles to the design of Biological science equipments
- Implementation of Biological science equipments
- Analysis of Biological science equipments
- Evaluation of Biological science equipments

Communicate effectively both orally and in writing and collaborate effectively in teams

- Conduct effective professional technical and non-technical conversations with co-workers and clients
- Communicate effectively in writing
- Work effectively in a team environment with people from other disciplines, throughout the problem solving process

Aims and objectives of teaching Biological science in Schools in different stages

It can be divided into three stages

1. Aims and objectives of teaching Biological science at Primary stage.
2. Aims and objectives of teaching Biological science at High School stage.
3. Aims and objectives of teaching Biological science at Higher Secondary stage.

1. Aims and Objectives of teaching Biology – Primary school

- Developing the habit of observation, classification and systematic way of thinking.

- To know about the characteristics of plants and animals and the benefits.
- To know the importance and the benefits of soil, air, water, sun, light, rain and warmth
- Developing the habit of observation, exploration, classification and systematic way of thinking.
- Developing the child's powers of manipulative, creative and inventive faculties.
- Arousing and maintaining interest in nature and in the Biological and social environment, arousing love for nature and its sources.
- know the ways and means of prevention and eradication of diseases to maintain good health
- Know the importance on cleanliness and formation of healthy habits.

2. Aims and Objectives of teaching Biology – High school

- Science should be taught as physics, chemistry and Biology.
- A disciplinary approach to science learning instead of general science would be more effective in providing the necessary scientific base to young people.
- Realizing scientific facts through experience
- To help students to appreciate and enjoy their environment
- To enable him to develop the scientific attitude.
- Developing the ability to reach generalisation and to apply them for solving every problem.

- To make scientific works through scientific exhibition and activities

3. Aims and Objectives of teaching Biology – Higher Secondary school

Biology is taught as a separate subject in higher secondary schools.

- To familiarize the pupil with the world in which he lives and to make him understand the impact of science on society so as to enable him adjust himself to his environment.
- To develop the scientific attitude
 - Followers to cause and effect relationship
 - Adopts planned procedure in solving a problem
 - Truthful in his experimentation and conclusion
- Fostering of creativity leading to innovations in science.
- To acquaint him with the 'scientific method' and to enable him to develop the scientific attitude.
- Developing interest in scientific hobbies.
- Ability to use basic laboratory tools and apparatus safely.

1.2.2 Difference between Aims and Objectives

S. No.	Aims	Objectives
1	Aims are long term goals	Objectives are short term goals

S. No.	Aims	Objectives
2	Aims are broad and common	Objectives are narrow and specific
3	Philosophy, Sociology is main source of aim	Psychology is the main source of objectives
4	They are difficult to achieve	they can be achieved conveniently
5	These can't be evaluated	They can be evaluated
6	School, Society and Nation is responsible to achieve them	Generally teacher is only responsible
7	Aims are non-observable	Objectives are observable behaviours
8	To achieve aims we have to go beyond curriculum	Objectives are achievable within the curriculum

1.3 Need and significance of teaching Biological Science

To teach any subject, it is the primary fact to know the teachers and students. What is the purpose of teaching and what are the need and uses of teaching.

Biological Science is a very important subject. The aims and objectives of Biological Science teaching have undergone an immense change in the computer era. In the growth and development of an individual and to apply knowledge and skills in the working place, we need

Biological Science. Any how the following points will tell us the need and importance of Biological science.

- Biological functioning in everyday life needs to be explored with the cause and consequences. Accordingly biology education is required to take these actions to benefit human life or to take precautionary measures.
- Biological science based research projects will develop their scientific knowledge of the students.
- Knowledge about natural environment, relation between plant and animals, living and non living, interrelation between plants, animals, their self defense mechanism, adaptations, nutrition, reproduction and evolution helps the human to lead a compatible life with nature.
- To help the students to get awareness about their environment.

Ideas, facts, experiences gained from projects and the results offer an everlasting knowledge to the students. These experience will help to solve problems in their life.

Significance of teaching Biological science

1. To understand the nature of Biological Science.

- To understand the scientific principles, laws and logic which is used in present or in future.
- To understand science in healthy and truth.
- To understand the world around us and the natural laws
- To understand the various parts of science and their relationship

2. To obtain thinking skills

When student sit in the class rooms or in laboratories to get Biological education knowledge, they obtain several skills.

- Obtain the problem solving skill
- Obtain drawing skills to be used in experiment
- Laboratory knowledge need in future

3. To obtain the observation skill

Only in Science subject five senses are used to find the truth. Concepts and principles are observed properly and find the truth with the help of experiments

4. To develop the scientific attitudes

Students get the following scientific attitudes while learning Biological Science

- Getting broad scientific attitudes
- To get the interest in Biological Science
- To obtain knowledge to solve the science problems
- Not to be believer of superstitious.
- To respect the views of others
- To be open minded

5. To get training in scientific methods

Students solve their day to day problems using the knowledge obtained in Scientific methods

Steps in Problem solving using scientific method

- Identify the problem

- Define the problem
- Collect required data to solve the problem
- Forming hypothesis
- Testing hypothesis
- Apply the results in various other situations

6. To create interest in Science

Teaching Science help students to participate in scientific activities with interest. Students take interest in participating scientific activities namely Science Club, Science Excursion, Field Trip and Science exhibition etc.

7. To know how Biological science is used for technological development

We cannot attain technological development without the help of Biological science. The things which we are using to live comfortable are made with the help of Biological science. It gives various methods to keep environment without pollution. New technologies are introduced with the help of Biological Science.

8. Training for better living

A science student should know the ways and means of prevention and eradication of diseases and to maintain good health. He should know how to adjust himself with his own domestic social environment and economic and cultural conditions.

9. To make our leisure time usefully

Biological science make our leisure time usually by maintaining vegetable gardens planting ornamental plants. honey extracting, pet animals etc. From these activities we understood the scientific concepts clearly and it also gives some remuneration. The aims and objectives of education should also give how to make our leisure of time usefully.

10. To prepare students for the future

Science helps to prepare students for their future. In Higher secondary stage, students should select the subjects according to their choice and keeping in mind their future. School should encourage students to select subject to study engineering, technical or communication branches for their future. Apart from physics and chemistry school should encourage students to take new subjects. Students who are studying Biological science in +2 stage will study these courses and make their future bright.

1.4 Nature of Biological Science

1. The World is understandable

Science presumes that the things and events in the universe occur in consistent patterns that are comprehensible through careful, systematic study. Scientists believe that through the use of the intellect, and with the aids of instruments that extend the senses, people can discover patterns in all of nature.

2. Scientific ideas are subject to change

Science is a process for producing knowledge. The process depends both on making careful observations of Phenomena and on inventing theories out of those

observations. Change in knowledge is inevitable because new observations may challenge prevailing theories.

3. Scientific knowledge is durable

Although scientists reject the notion of attaining absolute truth and accept some uncertainty as part of nature, most scientific knowledge is durable. The modification of ideas rather than their outright rejection, is the norm in science; powerful constructs tend to survive and grow more precise and to become widely accepted.

4. Science cannot provide complete answers to all questions

There are many matters that cannot usually be examined in a scientific way. There are for instance, beliefs that the existence of super natural powers. In other cases, a scientific approach that may be valid is likely to be rejected as irrelevant by people who hold to certain beliefs (such as miracles, fortune-telling, astrology and superstition).

5. Scientific Inquiry

Fundamentally, the various scientific disciplines are alike in their reliance on evidence, the use of hypothesis and theories, the kinds of logic used and much more. Scientists differ one another in what phenomena they investigate and how in they go about their work. Still the exchange of techniques, information and concepts goes on all the time among scientists and there are common understandings among them about what constitutes an investigation that is scientifically valid.

Scientific inquiry is not easily described. There is no fixed steps that scientists always follow. No one path that leads them unerringly to scientific knowledge. There are

certain features of science that leads to mode of inquiry. Those features are especially the work of professional scientists. Everyone can think scientifically about many matters of interest in every day life.

6. Science Demands Evidence

Scientific claims is settled by referring to observations of phenomena. Hence scientists concentrate on getting accurate data. Such evidence is obtained in Natural or in laboratory. To make their observations scientists use their own senses and instruments. To reliance on evidence, great value is placed on the development of better instruments and techniques of observations and the finding of anyone investigator or groups are usually checked by others.

7. Science is a Blend of logic and imagination

Scientific findings must confirm to the principle of logical reasoning that is testing the validity of findings by applying certain criteria of inference, demonstration and common sense.

Scientists do not work only with data and well developed theories. First they form tentative hypothesis about the way things may be. Such hypothesis are widely used in science for choosing what data to pay attention and what additional data to seek.

Scientific concepts do not emerge automatically from the data or from analysis alone. Inventing hypothesis or theories to image how the world works and then post to the test of reality is as creative as writing poetry, composing music etc. Sometimes discoveries in science are made unexpectedly. But knowledge and creative insight are usually required to recognize the meaning of the unexpected. Aspects of data

that have been ignored by one scientist may lead to new discoveries by another.

1.5 Scope of Biological science

Biological has a wider spectrum.

In recent years, Biological is studied under various interdisciplinary branches.

Example:

- Micro Biological
- Biotechnology
- Genetic Engineering
- Forensic science
- Agricultural Engineering
- Bio informatics
- Bio chemistry
- Horticulture
- Veterinary science
- Animal Husbandry
- Dairy
- Fisheries

In this technological world, scientific discoveries play an important place in day today life of human beings. To fulfill his basic needs he relies on plants and animals for his food and clothing. Biology plays an important role in major aspects like art, culture, entertainment, import and export. This can be an application value. Not only this, at times of environmental disorders, in the production of anti infectious drugs, fertilizers, insecticides, in plant and animal husbandry,

increasing the production of food, food preservation, and in the production of hybrids, Biology plays the key role.

1.6 Values of teaching Biological Science

Importance and functions of values

Values are general principles to regulate our day to day behaviour. They not only give direction to our behaviour but also ideals and objectives in themselves. Values deal with what ought to be. Our values are the basis of our judgements about what is desirable, proper, correct, important, worthwhile and good. It controls our disruptive behaviours; values enable individuals to feel that they are above us. Live with the following objective.

“Let Noble thoughts come to me from all directions”

Values of teaching Biological science

The real importance of the subject in our modern life can be quite obvious from its chief values which are described below.

1. Practical Value

The main objective of practical method is to explain the relationship between everyday life and Biological. Biological is an important lesson in the school curriculum. Our activities are controlled, designed and fashioned by Science. Without Science our existence would become impossible. It is very essential to have elementary knowledge for becoming useful, productive, responsible member of the community invention. We use the applications of science in every movement. The usefulness of science is evident in our daily activities. The food we eat, the shelter, transport, medicines, artificial fertilization techniques, population control measures,

fertilisers to increase the yield, Livestock breeding, hybrid seeds are possible only through science. A person becomes a good citizen only when he uses scientific knowledge with wisdom. Thus science is highly useful in everyday life.

2. Social values

Social purpose is to explain how Biology can be used to manage and maintain the rightful approach to the social and economic system.

People trained in various disciplines of Biological science are physicians, agricultural researchers, gardeners, scientists and veterinary doctors. They serve with the task of fulfilling the needs of society and for the development and progress of the country. This is the social values.

In many fields of Biology, scientists engage in research and their inventions are very useful to the society. Nobel prizes are awarded to their work.

3. Disciplinary Value

Disciplinary values deals with how Biological science is used to do any work with self confidence and discipline. The learning of Biological science involves some scientific disciplines and scientific attitudes which are transferable to our later life also. It involves self-expression, creativeness open mindedness, critical thinking and observation suspended judgment.

4. Cultural Value

Cultural values deals with how Biological science is used in the development of cultural and civilizational values of a nation. The role of Biological science in the development of modern civilization can be quite obvious just

by our comparison with our ancestors. Our present culture and advancement in our standard of living gives a clear cut picture of our cultural development and role of Biological science in this field for removing old traditional beliefs and superstitions.

5. Moral Values

Moral values are the standards of good and evil which governs an individual's behaviour. Science has more moral values. Science is the search for truth in a faithful manner. Promotion of science along with the growth of moral values is necessary for human development. Scientists strongly abhor fraud, error and pseudoscience while they value reliability, testability, accuracy, precision and simplicity. These values are generally derived from our experience in research. Good views of others are accepted by scientists. For the welfare of mankind scientists conduct research and invent new things. It teaches the pupil to be intellectually honest and truthful in our daily activities.

6. Aesthetic Value

Knowledge of Science develops in man a passion for truth thus he has a passion of beauty. Nature is store house of all beautiful things. Teaching of Biology is essential for developing Aesthetic sense in an individual.

Science has its own beauty since it is related with the beauty of nature, for his peace of mind and mental pleasure man depends on nature. Visiting the places like waterfalls, parks, sanctuaries, botanical gardens, zoological parks and growing ornamental plants, maintaining terrace garden, rearing pets animals, they enjoy the beauty of their work. Truth, Goodness and beauty are the theme of science. We become aware of the harmony of nature only when we possess basic scientific understandings.

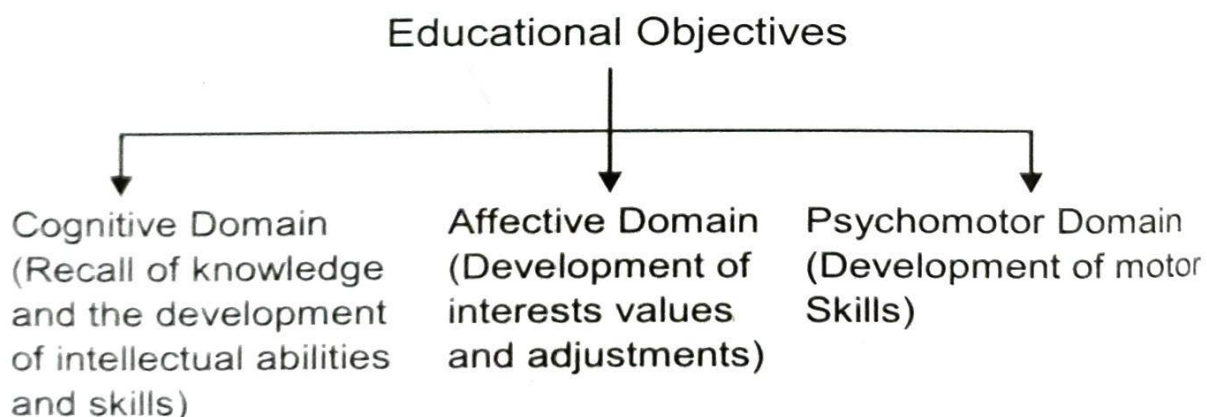
Conclusion :

So at the end, from the above mentioned values of Biological science, we come to this conclusion that Biological science has achieved a most important place in our daily life as well as in the modern establishment of the whole world.

1.7 Bloom's Taxonomy of educational objectives

To define understand and classify the educational objectives American educationalist Benjamin S. Bloom and his associates at the university of Chicago divide it into three domains i.e. cognitive, affective and psycho-motor and explains clearly to the teachers.

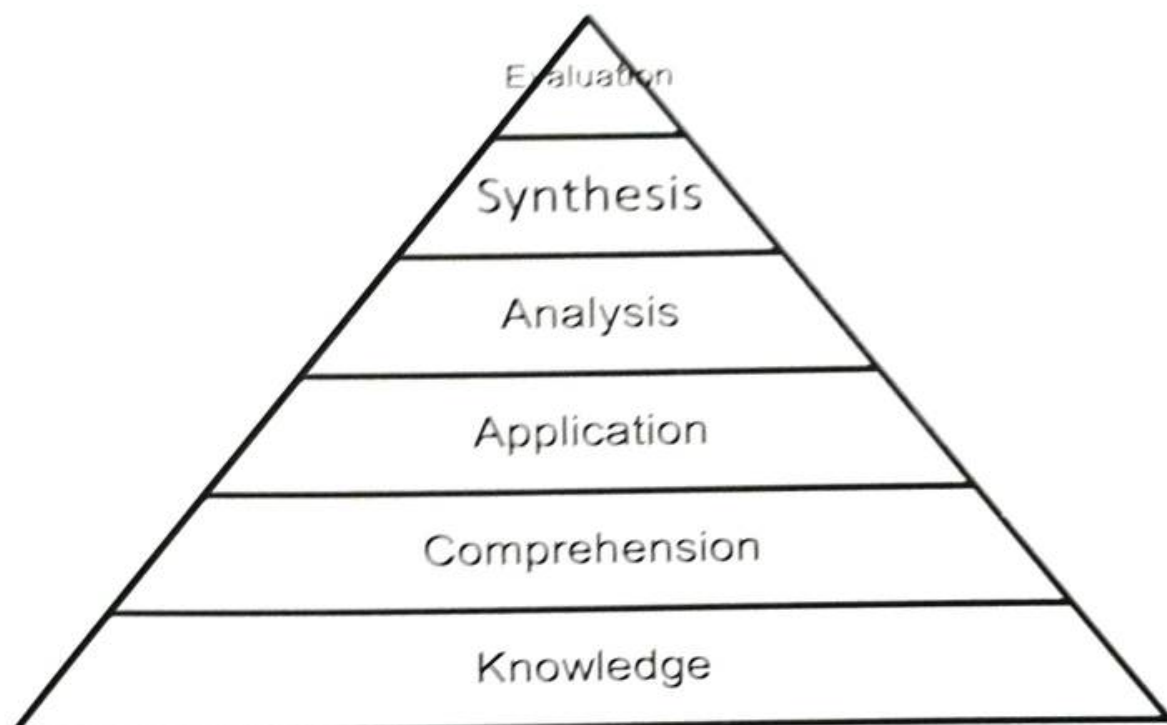
A brief account about these is given below.



In each domain, objectives are hierarchical order starting from easy to difficult. Hence this arrangement is called "Taxonomy of Educational objectives.

1.7.1 Cognitive domain

Cognitive domain includes those objectives which deal the recall or recognition of knowledge and the development of intellectual abilities and skills. This domain consists of six parts and arranged in a hierarchical order, in increasing levels of difficulty (from lower to upper)



Knowledge forms the basis for all cognitive objectives and all other objectives are not possible without knowledge. Similarly if one is able to apply knowledge in a new situations, it can be inferred that he has to achieve lower levels of objectives namely knowledge and comprehension clearly. Thus a higher level objective subsumes to lower ones.

While teaching biological science for Secondary and Higher Secondary students give priority to the first three objectives of cognitive domain namely knowledge, comprehension and application. The remaining three objectives Analysis, Synthesis and Evaluation may be trained to them according to their maturity of mind.

Knowledge

This objective is to remember and recall or recognize the already learnt information, acquire more and more knowledge.

Comprehension

Objective at this level requires understanding the knowledge without any doubt. He should have ability to arrange data in order.

Application

This objective requires students to apply already learnt knowledge learnt knowledge to new and unfamiliar situations. Think critically in depth.

Analysis

Objective at this level requires the learners, to attain the ability to break up a given communication into parts to understand it better.

Synthesis

Synthesis is just opposite to analysis. The ability of this level is to draw upon elements from various sources and put them together to produce a new structure. This objective gives divergent thinking and creativity.

Evaluation

This is a highest objective of cognitive domain. The ability of this level is to make judgements about something for given purpose. Judgment can be both quantitative and qualitative.

It helps teachers in the following ways:

- Helps to assess and judge the difficulty of assignments.
- It increase the accuracy of the lesson
- Simplifies action to ease personal learning
- Designed a combined assessment

- Helps for project based plans for learning
- Helps for group discussion

Instruction objects of Cognitive Domain

Parts of Cognitive Domain	Instructional Objectives
<u>Knowledge</u> Student knows the biological science symbols, concepts, definitions, formulas, statements etc.	<u>Student</u> 1. Recalls the biological science formulas, concepts and statements. 2. Recognises what he learned.
<u>Comprehension</u> Student understand the of biological science subject science symbols, concepts, definitions, formulas, statements	<u>Student</u> 1. Gives examples 2. Give reasons 3. Compares 4. Differentiates 5. classifies 6. relates 7. detect errors and rectify 8. using symbols 9. Explains 10. Evaluate the result
<u>Application</u> Using of biological science knowledge and comprehension to solve the problem in a new situation.	<u>Student</u> 1. Examine what is given in the problem? What is need? 2. Examine the sufficiency of the data. 3. Select appropriate formula 4. Solve the problem in an appropriate method. 5. Applying alternate method 6. Verifying the result.

Parts of Cognitive Domain	Instructional Objectives
<u>Analysis</u> Break up a given of biological science statement into small meaningful parts	<u>Student</u> 1. Understand the meaning of the hypothesis 2. From the given data draw conclusion 3. Identify motives / causes for the biological science concepts. 4. Discriminates, selects, illustrate biological science concepts.
<u>Synthesis</u> Putting elements from various sources and put them together to produce a new structure	<u>Student</u> 1. Design a method to perform a new task. 2. Revises the process to improve the outcome. 3. Compile, generate, create biological concepts. 4. Solves the problem 5. Develop the idea in advance.
<u>Evaluation</u> Evaluating Qualitative or quantitative judgement for biological science elements or methods.	<u>Student</u> 1. Evaluate the solution of a given problem 2. Select the most effective solution. 3. Explain and justify a new proposal 4. Assess / defend biological science concepts. 5. Review / Criticise biological science statements.

1.7.2. Affective Domain

The affective domain refers to emotional and attitudinal engagement with the subject matter while the cognitive domain refers to knowledge and intellectual skills related to the subject matter.

Affective domain includes the development of feelings of the educational and social attitudes and to increase the emotional objectives namely attitude, interest, values and appreciation. The different objectives under affective domain are Receiving, Responding, Valuing, organisation and characterisation.

1. Receiving:

This is the first step for the learner to learn what the teacher teaches. Listen attentively to a particular phenomena or stimuli.

2. Responding:

This is also called interest objective. Pupil is sufficiently motivated to attend actively. Interest in mathematics lends the student to select a particular vocation throughout his life. Interest can be developed in solving puzzles, reading mathematical books and journals etc.

3. Appreciation :

Student appreciates Mathematics for its use in daily life and how it helps to other subjects.

4. Organisation :

This objective brings together different values, resolving conflicts among them, starting to build an internally consistent value system.

5. Valuing:

It means individual's own valuing or assessment of thing, phenomena or behaviour. It is also called attitude. Development of proper attitude in mathematics is one of the major objectives of teaching mathematics.

Instructional Objectives of Affective Domain

Parts of Affective Domain	Instructional Objectives
<u>Receiving</u> Listen to others with respect. Listen for and remember the newly introduced content.	<u>Student</u> 1. Identifies 2. Asks 3. Understand 4. Points out errors boldly. 5. Listen to others attentively.
<u>Responding</u> Participate in class discussions. Gives a presentation, questions new ideas, concepts etc.	<u>Student</u> 1. Reads books of science 2. Solves science puzzles 3. Writes activities on science for school magazine. 4. Gives short cuts for solving problems. 5. Does problems apart from syllabus. 6. Gains science knowledge more than the syllabus. 7. Participates in science club activities. 8. Participate in class discussions.
<u>Appreciation</u> Valuing to a particular object, phenomenon or behavior ranges from simple acceptance to the more complex state of commitment.	<u>Student</u> 1. Solving the problems of other subjects using science. 2. Appreciates how science is useful in his daily life.

Parts of Affective Domain	Instructional Objectives
<u>Organisation</u> Recognise the need for balance between freedom and responsible behavior. Explains the role of systematic planning in solving problems.	<u>Student</u> 1. Recognize the need for balance between freedom and responsible behavior 2. Explain the role of systematic planning in solving the problems. 3. It becomes habit to think logically.
<u>Valuing</u> Has a value system that controls his behavior. Uses an objective approach in problem solving. Revises judgments and changes behavior in the light of new evidence.	<u>Student</u> 1. Shows self reliance when works independently 2. Co-operates in group activities. 3. Revises judgements and changes behavior in light of new evidence. 4. Values people for what they are not how they look. 5. Accepted statements when logically proved.

1.7.3 Psychomotor Domain

Elizebath Simpson built her taxonomy on the work of Bloom and others. Simpson's Psychomotor domain is comprised of utilizing motor skills and coordinating them. Simpson's taxonomy has a focus toward the progression of mastery of a skill from observation to invention.

The psychomotor domain includes Physical movement, coordination and use of the motor-skill areas. Development of these skills requires practice and is measured in terms of speed, precision, procedures or techniques in execution. The seven major categories are perception, set, Guided Responses mechanism, complex over to Response, Adaptation and Origination.

The behaviour and development of every child is depends on the above three domains. So while teaching, the teachers teach the subject matter on the basis of the above three domains.

1. Perception

It means observing the act of the teacher.

2. Set

Readiness to act. It includes mental, physical and emotional sets

3. Guided Response

Following directions of the teacher and differentiating among various movements. This is the early stage in learning complex skill.

4. Mechanism

This is the intermediate step in learning a complex skill. Learned responses have become habitual and the movements can be performed with some confidence and proficiency.

5. Complex Overt Response

This is the highest level of proficiency and it becomes natural and automatic. Proficiency is indicated by a quick, accurate and highly coordinated performance, requiring a minimum energy.

6. Adaptation

Skills are well developed and the individual can modify movement patterns to fit special requirements

7. Origination

Creating new movement patterns to fit a particular situation or specific problem. Learning outcomes emphasize creativity based upon highly developed skills.

Instructional Objectives of Psychomotor Domain

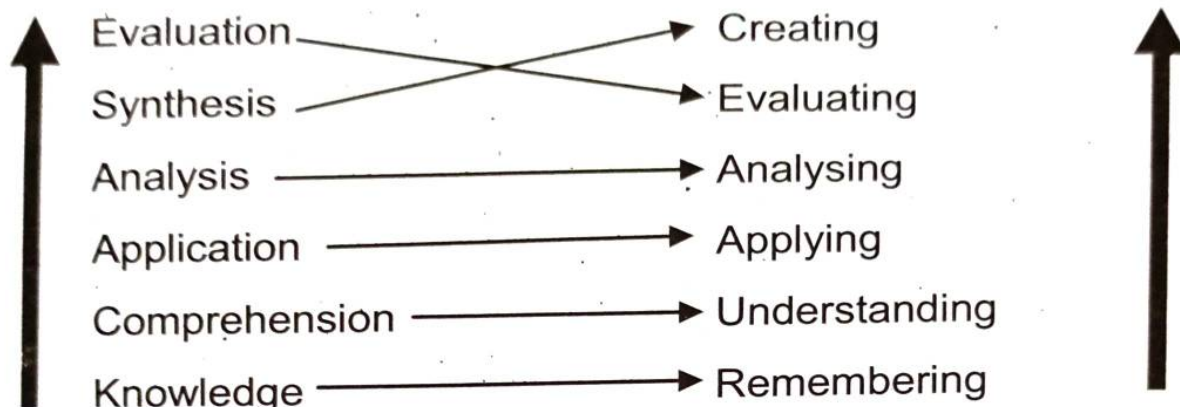
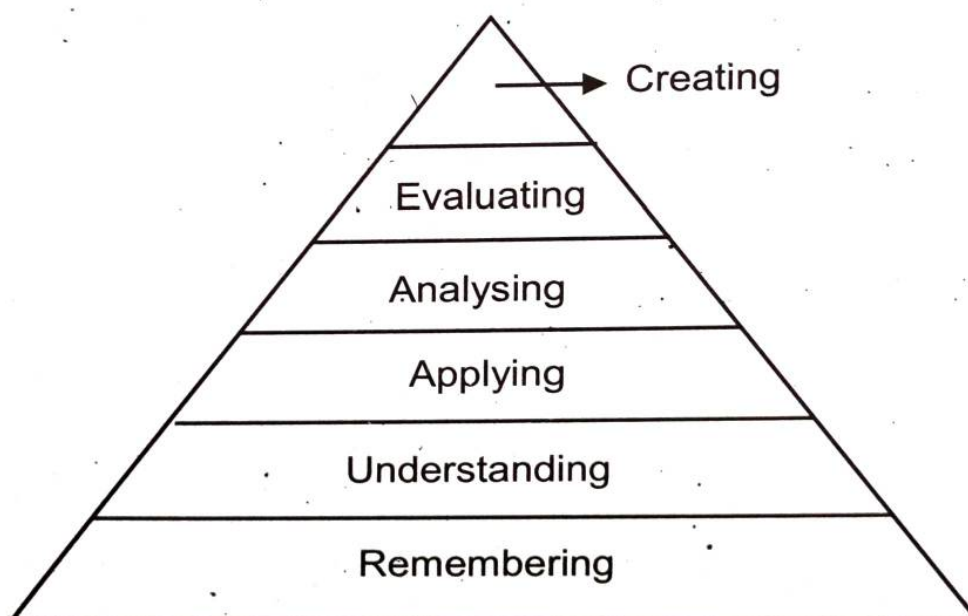
Parts of Psychomotor Domain	Instructional Objectives
Perception Perception is the most basic level to process sensory information that guide motor activity	Student 1. Detects non verbal communication cues 2. Estimate where a ball will land after it is thrown and then moving to the correct location to catch the ball. 3. Observing the act of the teacher.
Set Mentally, emotionally and physically ready to act.	Student 1. Repeat the act of the teacher. 2. Acts upon a sequence of steps to solve a problem. 3. Shows desire to learn a new process.
Guided Response Imitates and practices skills often in discrete steps	Student 1. follows 2. react 3. response 4. copies 5. performs
Mechanism Performs acts with increasing efficiency, co-ordinate and proficiency.	Student 1. Drives a cycle 2. assembles 3. dismantles 4. measures 5. mixes

Parts of Psychomotor Domain	Instructional Objectives
Complex Overt Response Motor activity performs automatically.	Student 1. Operates a Microscope quickly and accurately 2. assembles 3. builds 4. measures 5. organizes
Adaptation Adapts skill sets to meet a problem situation.	Student 1. Responds effectively to unexpected situations. 2. Modifies method to meet the needs. 3. Perform a task with a new method.
Origination Create new patterns for specific situations.	Student 1. Constructs a new theory. 2. designs 3. creates 4. builds 5. composes

The behaviour and development of every child is depends on the above three domains. So while teaching, the teachers teach the subject matter on the basis of the above three domains.

1.8 Revised Bloom's Taxonomy

In 1990 Loris Anderson former student of Bloom revised the taxonomy and made a number of changes.

Bloom's Taxonomy**Revised Bloom's Taxonomy****Revised Bloom's Taxonomy****Remembering:**

The knowledge category was renamed as remembering. Knowledge is an outcome or product of thinking and not a form of thinking. Hence the word knowledge was inappropriate to describe a category of thinking and was replaced with the word remembering.

Remembering is recalling information. This objective is to recognizing, listing, describing, naming, finding already learnt information.

Understanding:

Objective at this level requires learners to understand the knowledge without any doubt. He should have knowledge to interpret, explain, classify and summarize.

Applying:

Objective at this level requires the learners to apply information in another familiar situation and executing in a best way.

Analysing:

Objective at this level requires the learners to attain the ability to break up a given information into parts to explore understanding and relationships between them.

Evaluating:

The ability to make judgements about a decision or course of action. Judgement can be both quantitative or qualitative.

Creating:

By putting elements together from various sources to form the ability to generate new ideas, products or ways of viewing thing in a different way. Designing, planning, producing, inventing new things and objects is the objective at this level.

Potential activities of cognitive domain**Remembering:**

1. Make a list of the main events of today.
2. Write parts of human body.

Understanding:

1. Retell the story in your own words.
2. Prepare a flow chart to illustrate the sequence of events.

Applying:

1. Construct a model to demonstrate how it works.

2. Make a clay model to explain a biological concept.

Analyzing:

1. Design a questionnaire to gather information.
2. Construct a graph to illustrate given information.

Evaluating:

1. Conduct a debate about an issue of
2. Prepare a case to present your view about.....

Creating:

1. Invent a machine to do a specific task.
2. Design a magazine cover for sports day function.

1.9. Writing instructional objectives for different Biological Topics

To write objectives for a particular mathematics topic, begin with a action verb while writing objectives. Do not use verbs such as learn, understand, Discuss, explain etc, as these do not specify exactly how students will demonstrate their knowledge. use specific action verbs that will demonstrate their exactly what the student will be doing during the lesson.

Remember that your objectives must be observable. Can you see a student determine? Can you see a student understand? Do not use unobservable behaviors as objectives.

Class - 8 Genetics and Evolution

Learning objectives:

At the end of this Lesson students will be able to

1. tells about genetics, evolution and its properties
2. explain about the transmission of characters from generation to generation

3. using his knowledge and understanding the students acquires the ability to solve doubts.
4. They are able to find out the difference between the twins in their schools.
5. approaches the basic techniques of genetic engineering with research mentality.
6. willing to solve the questions given to him.
7. able to explain that why human traits are transmitted.
8. able to write the typographical details in coding form.
9. appreciates the structure of gene therapy.
10. writes articles on the genes and traditional evolution in his school annual day.

Class - 10 Viral diseases

Learning objectives:

At the end of this Lesson students will be able to

1. tells about the virus and the disease caused by the virus.
2. explains about the virus and the disease caused by the virus.
3. using his knowledge and understanding the students acquires the ability to solve questions in a new environment.
4. acquires the ability to control the viral diseases.
5. approaches the various types of viruses with research mentality
6. tries to solve the problem with the practical solution.
7. tells about the types of virus and their types based on DNA and RNA.
8. tells the examples for viruses

9. perform the practicals fastly and accurately.
10. finds out the symptoms of viral diseases.

Class - 8 Movement of animals

Learning objectives:

At the end of this Lesson students will be able to

1. tells about the movement of animals and functions
2. explains about the movement of animals and functions.
3. using his knowledge and understanding the students acquires the ability to solve doubts.
4. able to examine the body movement of the different animals and characteristics of the movement.
5. able to improve his practical knowledge.
6. able to solve the problem given to him.
7. differentiate the body movement of the birds during down stroke, up stroke and landing.
8. able to explain the movement of fish, snake and worms.
9. examines the various components in the given problem.
10. able to find a solution for the problem.

Class - 8 Body movements

Learning objectives:

At the end of this Lesson students will be able to

1. tells about the Human body and its movements.
2. tells that human body perform all the activities with the help of the movement of bones and muscles.
3. using his knowledge and understanding the students acquires the ability to solve questions in a new environment.

4. able to draw the skeletal system of man.
5. draw the skeleton model with the research attitude
6. draws the structure of human skeleton with interest.
7. explains what was given in the question and what was asked?
8. tells about the movement of the joints.
9. tells about the structure of the skeletal system from the laboratory.
10. tells about the exact number of bones present in the skeletal system.

Class - 9 Animal kingdom

Learning objectives:

At the end of this Lesson students will be able to

1. tells about the classification of the animal kingdom.
2. tells about the classification of the animal kingdom.
3. using his knowledge and understanding the students acquires the ability to solve questions in a new environment.
4. able to set up the sages in classification and the basic units.
5. able to approach the classification with the scientific attitude.
6. able to solve the problem given to him.
7. able to compare each step in the classification.
8. able to find out the solution for the problem in an appropriate manner.
9. examines the various components to the given problem.
10. able to solve the problem and find the answer.

1.10 Inter-relation among the Domains

Learning helps develop an individual's attitude as well as encourage the acquisition of new knowledge and new skills.

Learning can generally be categorized into three domains: Cognitive, affective and psychomotor. Learning is not an event. It is a process. It is the continual growth and change in the brain's architecture that results from the many ways. We take in information, process it, connect it, catalogue it and use it.

Cognitive Development

Cognitive development refers to the way a child grows intellectually. Assessing for academic skills, reasoning, and critical thinking abilities, and the acquisition of knowledge or learning. Difficulties interacting with others as a result of problems and social or emotional development will impede a child's ability to focus. Problems or delays in physical development might impact a child cognitively.

Affective Domain

The affective domain includes the feelings, emotions and attitudes of the individuals. The categories of affective domain include receiving phenomena; responding to phenomena; valuing; organization and characterization. The sub domain of receiving phenomena creates the awareness of feelings and emotions as well as the ability to utilize selected attention. This can include listening attentively to lessons in class. The next sub domain of responding to phenomena involves active participation of the learner in class or during group discussion. The ability of the student to prioritize a value over another and create a unique value

system is known as organization. This can be assessed with the need to value one's academic work as against their social relationships. This helps to control the behavior of the individual.

Psychomotor domain

This includes utilizing motor skills and the ability to coordinate them. The sub domains are perception; set; guided response; mechanism; complex overt response; adaptation; and origination. Perception involves the ability to apply sensory information to motor activity. for eg. Practicing a series of exercises for scoring high marks. Set, involves the readiness to act upon a series of challenges to overcome them. It includes the ability to imitate a displayed behavior or utilize a trial and error method to resolve a situation. Mechanism involves the ability to convert learned responses into habitual actions with proficiency and confidence. A student having an increased typing speed when using a computer is an example for complex overt response. Adaptability is an integral part of the domain which exhibits the ability to modify learned skills to meet special events – making a working model. Origination also involves creating new movement patterns for a specific situation.

Cognitive is for mental skills, affective is for growth in feelings or emotional areas while psychomotor is manual or physical skills.

The learning domains can be incorporated, while designing the course outcomes of all the courses in a program. We can learn mental skills, develop our attitudes and acquire new physical skills as we perform the activities of our daily living.

Research in the field establishes the significant relation among three domain of learning.

A lesson developed by a teacher requires the inclusion of all three domains in constructing learning tasks for students. The diversity in such learning tasks helps creates a comparatively well-rounded learning experience that meets a number of learning styles and learning modalities. It is important for teachers to adopt a teaching strategy that combines three domains of learning to enable teaching and learning to be considered as effective. Apply the best and suitable delivery techniques that would impact positively on the Cognitive, Affective and Psychomotor domains of the students. The interaction of all three domains is needed for meaningful educational experiences.

For example while teaching Biological Science, New knowledge is acquired in Cognitive Domain. Using motor skills we teach how do the best in practical. With the Affective Domain components we give motivation, praise to students to bring interest in the subject matter. In the same way we relate two or three Domains and perform teaching - learning activities and give meaningful learning experiences.

1.11. Correlation of Biology with other subjects

Science cannot be taught in isolation. All the branches of science are interdependent. A number of facts and principles are common to various science subjects. A teacher should have sufficient knowledge of other sciences so as to bring about integration of subjects.

1. Biology and language

Biology students are usually weak in their expression. In order to make the science students able to express their thoughts in clear, concise, correct and attractive, the Biology teacher and the language teacher must take up a joint responsibility for the cultivation of a good style in answering essay type questions. The language teachers can ask the students to write an essay on some invention or on the life history of a scientist. The Historical events the biographies of scientists provide an excellent material for reading.

2. Biology and History

In fact there is no direct relation between. History and Biology as History is related with past time and biology is inter-related as the sequential form of History. All the inventions are discovered are found in History and History tells about the odd circumstances in which scientist discovered the things which helps students to learn to live in all situations for the welfare of the society. The history of scientific inventions and discoveries provides useful background for the teaching of history.

3. Biology and Mathematics

Biology is incomplete without mathematics. Mathematics have given a sound footing to the scientific law. The knowledge of mathematics is important for the real understanding of science. Many mathematical topics namely decimals, proportion, inverse ratio, equations, graphs are used in Biology.

Appropriate statistical methods are used to analyze multi-level biological data. The teacher of Biology and of

Mathematics work in cooperation to bring about correlating and coordination of the two subjects.

4. Biology and Geography

Geography is too much related to Biology that it has now been regarded as a branch of science. The study of soil, climate, distribution of plants and animals etc. brings the two subjects close to. The topics like pressure, humidity, the effect of climate and other facts on the plants, animals etc are common to both Biology and geography. Hence the biology teacher and the geography teacher should take cooperative efforts about a close integration of both the subjects.

5. Biology and Civics

Biology and Civics both tends to make a student efficient, ethical, dutiful and perfect citizen. Science develops feelings of social responsibilities among them. To be a good and dutiful citizen it is essential to know how to keep ourselves healthy, how to control pollution etc. These informations are mainly provided by the civics. The topics like food, weather, habitat and clothing are the common in biology and civics. Students study the above topics to live their life comfortably. Need of food, basic physiology, Hygiene, cleanliness are taught in biology and civics.

6. Biology and Agriculture

Biology and Agriculture are directly inter related to each other. Through Biology student gets information about structure and composition of soil, fertilizers, disease of plants, which help the farmer to grow better crop and get better yield.

7. Biology and Physical Science

Green revolution occurred due to the use of suitable chemicals on a large scale. Use of artificial fertilizers, pesticides, advance method of sowing and machineries to harvest the crops etc are done with the help of Physical Science.

Body of all living organisms made up of organic and inorganic compounds dissolved in water. All the metabolic activities inside the body of living beings involve chemical reactions and chemical changes.

Biological instruments such as microscopes, x-rays, ECG, Body scanner etc. are based on the principles of physics. Study on the structure and function of human eye require knowledge of optics.

Biology with other New subjects

New Subjects like Bio Physics (Biology and Physics), Bio Chemistry (Biology and Chemistry), Bio Mathematics (Biology and Mathematics), Bio Geography (Biology and Geography) are introduced in the curriculum.

Uses of correlation

1. Mind perceives knowledge as a whole. All the subjects aim to develop the intellectual power of the students as a whole. Therefore, it becomes necessary to correlate one subject with another. No subject can be taught in isolation.
2. Correlation of a subject with daily life helps to make the knowledge broad-based and makes knowledge useful. The aim of education "to achieve the development of an

all-round personality", cannot be done by teaching only a few subjects in isolation.

3. It strengthens skills that students encounter in one content area but also practice in another, leading to mastery of those skills.
4. It develops the mental abilities like imagination power, logical thinking and analytical power of students.
5. It makes the lesson easy and clear and develops various human and social qualities such as co-operation, citizenship etc.
6. Correlation of subject with daily life is of the utmost importance in order to create interest in the subject.
7. It makes learning concrete and permanent.
8. It helps to lighten the burden of curriculum.

REVIEW QUESTIONS

1. Explain the meaning, nature and scope of Biological Science.
2. What are the important Aims of teaching Biological Science. in Schools?
3. Write the different values of Biological Science.
4. Explain the Cognitive Domain and write its Instructional Objectives.
5. Explain the Affective Domain and write its Instructional Objectives.
6. Explain the Psychomotor Domain and write its Instructional Objectives.

7. Explain Revised Bloom's of Taxonomy.
8. How learning domains can be inter-related in teaching of Biological Science.
9. Explain how Biological Science is correlated with other subjects.
10. Write the need and significance of teaching Biological Science in School.